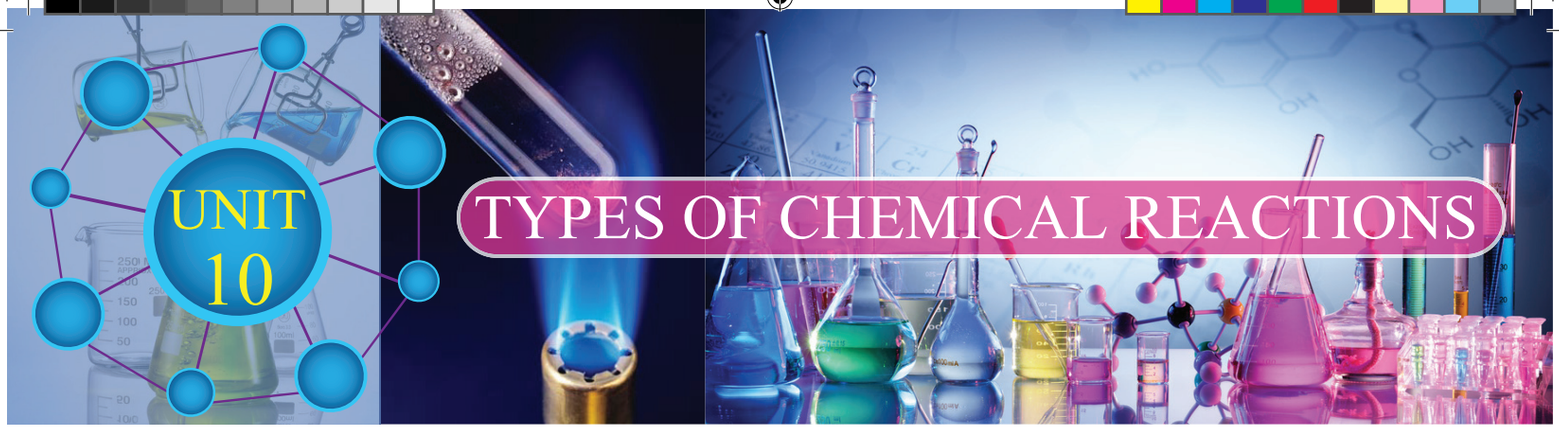


UNIT 10

TYPES OF CHEMICAL REACTIONS

Learning Objectives

T15NB

After completing this lesson learners will be able to

- infer different types of chemical reaction.
- acquire knowledge about combination reaction and skill to perform a combination reaction using quick lime and water.
- identify and differentiate between reversible and irreversible reactions.
- explain the reversible reaction occurring at the equilibrium state.
- list and explain characteristics of equilibrium state.
- define rate of reaction.
- discuss the dependence of rate of reactions on concentration, temperature and catalyst.
- define pH.
- correlate the concentration of hydrogen ions and pH with neutral, acidic and basic nature of aqueous solutions.
- recognize the importance of pH in everyday life.
- explain the term ionic product of water.

INTRODUCTION

As you know from your earlier studies, a chemical reaction involves breaking of old chemical bonds and formation of new chemical bonds. This change may happen spontaneously or it may be facilitated by external forces or energy. Chemistry is all about chemical reactions. In your day to day life, you could observe many chemical reactions. A clear understanding of these reactions is essential in order to manipulate them for the sake of human life and environment. So, chemistry mainly focuses on chemical reactions. Let us try to find the answer for the following questions:

- You need energy to play, walk, run or to perform various physical activities. Where do you get the energy from?
- How do plants grow and get their food?
- How does a car move using fuel?
- Why does iron rust on its exposure to water or air?

You get energy from the digestion of the food you eat. Plants grow by absorbing nutrients from the Earth and get their food by photosynthesis. The combustion of a fuel makes the car to move. Oxidation of iron causes rusting. So, all these processes are chemical changes i.e. the materials, which undergo changes are converted into some other new materials. For example, by burning petrol, the hydrocarbons present in it are converted into carbon dioxide and water. In this chapter, let us discuss the nature and types of chemical reactions.

10th_Science_Unit-10.indd 137 12/3/2022 2:55:18 PM